

50 SQL interview questions

Here are **50 SQL interview questions** that are commonly asked for both **freshers** and **experienced professionals**. I've organized them by difficulty level.

SQL Basics (Freshers)

1. What is SQL?

2. What are the different types of SQL commands?

- DDL
- DML
- DCL
- TCL
- DQL

3. What is the difference between DELETE, TRUNCATE, and DROP?

4. What is a Primary Key?

5. What is a Foreign Key?

6. What is a Unique Key?

7. What is the difference between Primary Key and Unique Key?

8. What is NOT NULL constraint?

9. What is the difference between CHAR and VARCHAR?

10. What are aggregate functions?

Examples:

- COUNT()
- SUM()

- AVG()
 - MAX()
 - MIN()
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Queries & Filtering

- 11. What is the WHERE clause?**
- 12. What is the difference between WHERE and HAVING?**
- 13. What is ORDER BY?**
- 14. What is GROUP BY?**
- 15. How do you find duplicate records in a table?**
- 16. How do you delete duplicate records?**
- 17. What is DISTINCT?**
- 18. Difference between IN and EXISTS?**
- 19. Difference between BETWEEN and IN?**
- 20. How do you retrieve the top 5 highest salaries?**

Example:

```
SELECT *
```

```
FROM Employee
```

```
ORDER BY Salary DESC
```

```
LIMIT 5;
```

Joins (Very Important)

- 21. What are SQL Joins?**
- 22. Difference between INNER JOIN and OUTER JOIN?**
- 23. Explain LEFT JOIN.**
- 24. Explain RIGHT JOIN.**

- 25. Explain FULL OUTER JOIN.**
 - 26. What is SELF JOIN?**
 - 27. What is CROSS JOIN?**
 - 28. When would you use SELF JOIN in real projects?**
 - 29. Difference between JOIN and UNION?**
 - 30. Find employees who do not have a manager.**
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Subqueries & CTEs

- 31. What is a Subquery?**
- 32. Difference between Correlated and Non-Correlated Subquery?**
- 33. What is a CTE (Common Table Expression)?**

Example:

```
WITH Sales_CTE AS
```

```
(
```

```
    SELECT *
```

```
    FROM Sales
```

```
)
```

```
SELECT * FROM Sales_CTE;
```

- 34. Advantages of CTE over Subqueries?**
 - 35. What is a Recursive CTE?**
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Window Functions (Frequently Asked)

- 36. What is ROW_NUMBER()?**
- 37. Difference between ROW_NUMBER(), RANK(), and DENSE_RANK()?**
- 38. How do you find the 2nd highest salary?**

```
SELECT MAX(Salary)
FROM Employee
WHERE Salary <
(
    SELECT MAX(Salary)
    FROM Employee
);
```

39. What are Window Functions?

40. Explain PARTITION BY.

Intermediate to Advanced SQL

41. What are Indexes?

42. Difference between Clustered and Non-Clustered Index?

43. What is Normalization?

44. What are the Normal Forms?

- 1NF
- 2NF
- 3NF
- BCNF

45. What is Denormalization?

46. What is a View?

47. Difference between View and Materialized View?

48. What is a Stored Procedure?

49. What is a Trigger?

50. What is ACID Property in SQL?

- Atomicity
 - Consistency
 - Isolation
 - Durability
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Frequently Asked SQL Coding Questions

1. Find the 2nd highest salary.
 2. Find the nth highest salary.
 3. Find duplicate records.
 4. Remove duplicate records.
 5. Find employees earning more than their manager.
 6. Find top 3 salaries department-wise.
 7. Find departments with no employees.
 8. Find customers who never ordered.
 9. Find the highest salary in each department.
 10. Calculate running total using window functions.
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Questions Commonly Asked to Experienced Professionals (2–5 Years)

Scenario-Based Questions

1. How would you optimize a slow query?
2. How do indexes improve performance?
3. What causes deadlocks?
4. How do you handle millions of records efficiently?
5. Explain query execution plans.
6. Difference between OLTP and OLAP?
7. Explain partitioning.

8. What is database sharding?
9. How would you design a sales database?
10. Explain transaction isolation levels.

If you're preparing for Data Analyst, SQL Developer, Business Analyst, Data Engineer, or BI roles, these 50 questions cover about 80–90% of what is commonly asked in interviews.